

AI StudyFlow

AI-Powered Student Productivity & Study Assistant

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Key Idea

- An AI-based study productivity application
- Helps students plan and organize their studies
- Generates personalized study tasks and schedules
- Provides AI quizzes and an AI study assistant

Technology Stack: Python | Streamlit | Groq AI | JSON



Introduction & Objective

Introduction

StudyFlow AI is an interactive web application designed to help college students manage their academic activities using Artificial Intelligence.

Objectives

- Create personalized study plans
- Organize study tasks and activities
- Manage available study time
- Generate AI-based quizzes
- Provide explanations for difficult concepts
- Track task completion and productivity

The application is specifically designed to keep study plans practical and avoid unnecessary workload.

Technologies & Tools Used



Programming Language

- Python

Framework

- Streamlit
- Used to create the interactive web interface
- Provides buttons, forms, menus, metrics and progress bars



AI Technology

- Groq API
- AI model used: openai/gpt-oss-120b



Other Libraries

- python-dotenv – environment/API key management
- datetime – dates and time
- json – processing AI-generated JSON
- os – environment variables
- re – regular-expression support

These imports and the Groq client setup are present at the beginning of the project.

System Architecture & AI Integration

Basic Working

1. Student Input
2. Streamlit Interface
3. StudyFlow AI Application
4. Groq AI Model
5. AI-Generated Response
6. Tasks / Activities / Schedule / Quiz / Explanation

AI Integration

The application uses a common `ask_groq()` function to send prompts to the Groq model and receive AI responses.

It supports JSON mode when structured AI output is required, such as productivity plans and quizzes.

Dashboard & User Interface

Dashboard Features

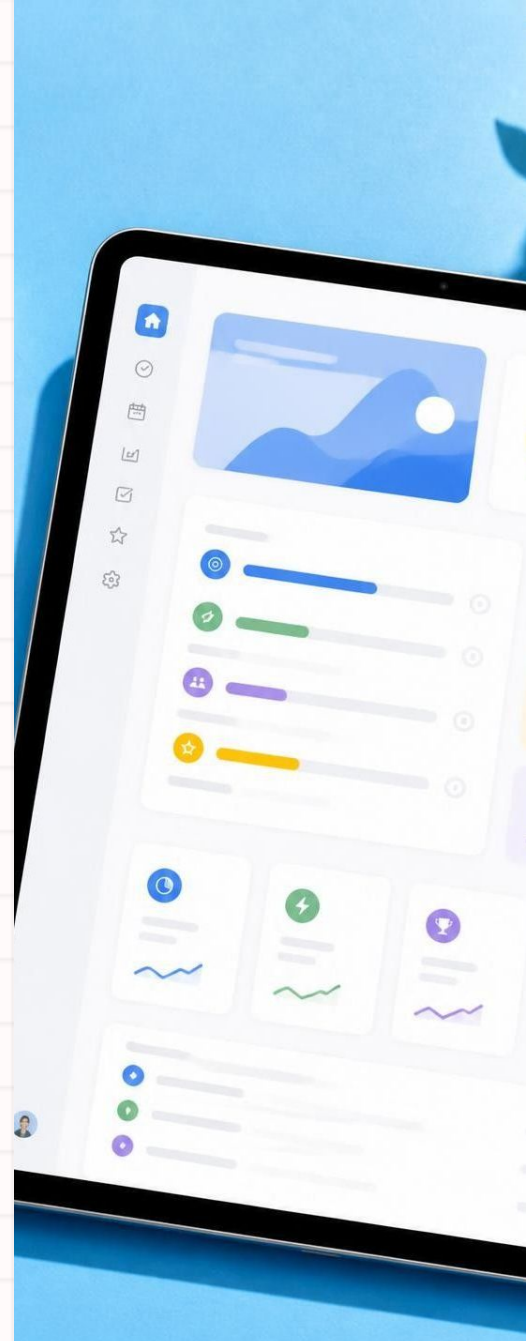
The Dashboard provides an overview of student productivity.

- Total Tasks
- Completed Tasks
- Pending Tasks
- Total Activities
- Productivity Progress
- AI feature overview

Navigation Menu

1. Dashboard
2. AI Productivity
3. Tasks
4. Activities
5. AI Quiz
6. AI Assistant

The dashboard calculates task completion percentage and displays it using a progress bar.



AI Productivity Generator

Purpose

The AI Productivity module creates a personalized study plan based on the student's requirements.

User Inputs

- Subject
- Topic
- Goal
- Difficulty
- Available study hours
- Deadline

Available Goals

- Exam Preparation
- Learning Concept
- Revision
- Assignment
- Project
- Interview Preparation

AI Generates

- Study Tasks
- Study Activities
- Time Schedule
- Productivity Tips

The AI is instructed to keep the total study time within the available hours and

Task & Activity Management

Task Management

AI-generated tasks are stored with:



Task name



Description



Priority



Duration



Category



Deadline



Completion status

Study Activities

- Activity/topic
- Type
- Duration
- AI-generated label
- Creation time

This allows students to see their planned academic work separately from their overall task list.

AI Quiz Module

AI Quiz Features

Students can select:

- Subject
- Topic
- Difficulty
- Number of questions

Quiz Generation

- Multiple-choice questions
- Four options per question
- Correct answer
- Explanation

Result Evaluation

- Score is calculated
- Percentage is displayed
- Progress bar shows performance
- Correct answers are reviewed
- Explanations are provided

The code validates that the AI returns the requested number of questions and exactly four options for each question.

AI Study Assistant & Session State

AI Study Assistant

Students can type academic questions and receive AI explanations.

Response Includes

1. Simple explanation
2. Example
3. Important points

Quick Topics

- Python
- DBMS
- Artificial Intelligence

Session State

The application uses Streamlit session state to maintain:

- Tasks
- Activities
- Generated plan
- Quiz
- Quiz score
- Quiz submission status
- Assistant question
- Assistant answer

This allows information to remain available while the user

Conclusion & Future Scope

Conclusion

StudyFlow AI combines Artificial Intelligence and productivity management into one student-focused application.

Main Benefits

- Personalized study planning
- Better time organization
- Automated task generation
- AI-powered quizzes
- Instant academic assistance
- Productivity tracking
- User-friendly interface

Future Scope

- User login and profiles
- Database storage
- Persistent task history
- Calendar integration
- Notifications and reminders
- More subjects and learning resources
- Progress analytics and reports

Final Message

“Learn Smarter. Plan Better. Achieve More.”

The current application footer identifies the project as “Powered by Groq + Streamlit.”

